# Build a prediction one coordinate at a time

**GUIDED NOTES / use with the projected diagram** Name _____

## A SAY THE WORD

### Reference frame / ES: sistema de referencia / PT: referencial

Keep the origin and axes fixed.

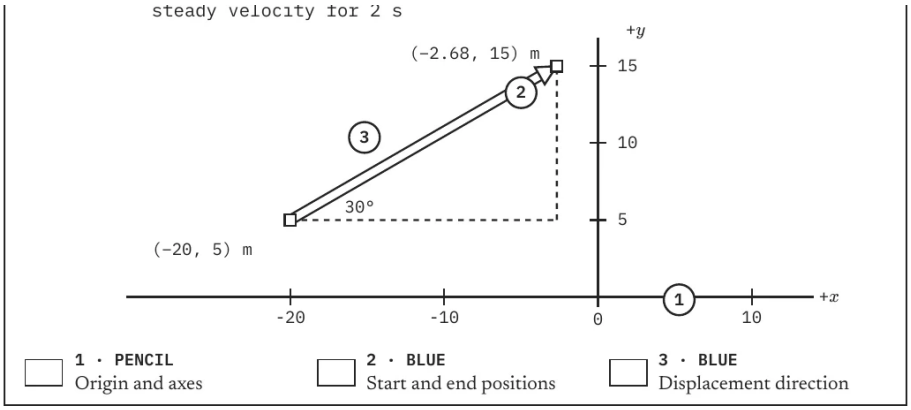
Also on your note page: Components, Units, Limit.

## B READ THE RELATIONSHIP

### Advance x and y independently using vx and vy in the same frame.

As you read, find: origin · direction · assumption.

## C TRACE THE ORIGINAL FIGURE




Use the numbered key. Trace one feature. Tell your partner what it means.

## D FOLLOW THE WORKED EXAMPLE

### A probe starts at (−20, 5) m and moves at 10 m/s , 30° from + x toward + y , for 2 s . Find its new coordinates.

**Given** x 0 = −20 m ; y 0 = 5 m v = 10 m/s ; θ = 30° t = 2 s ; steady v

**Rule** Resolve v : v x = 8.66 m/s v y = 5 m/s

**Work** x f = −20 + 8.66(2) y f = 5 + 5(2)

**Answer** (−2.68, 15) m Final x is still negative.

## E SAVE FOR THE FINAL MEETING

For fictional XJ, name the frame, units and motion assumptions. Fixed Earth and steady velocity are classroom simplifications.

# Read the physics: Reference frame

SUPPORTED READING / read one part, then point to the evidence Name _____

## 01 PICTURE IT

Name a reference point, mark the origin and choose axes. Record the starting position and the time. Include a unit with each value.

## 02 HOW IT WORKS

If velocity is steady, add velocity times elapsed time to the starting coordinate. In two dimensions, do that once for x and once for y. Use the matching component each time.

## 03 WHAT TO CHECK

Check whether the final position is on the expected side of zero. Say which motion assumption could fail when a new sighting arrives.

## 04 TRY THE EXAMPLE FROM YOUR NOTES

A probe starts at (−20, 5) m and moves at 10 m/s , 30° from + x toward + y , for 2 s . Find its new coordinates. (−2.68, 15) m Final x is still negative.

**Model check: Advance x and y independently using vx and vy in the same frame.**

## 05 CONNECT TO XJ

For fictional XJ, name the frame, units and motion assumptions. Fixed Earth and steady velocity are classroom simplifications.

SAY IT BEFORE YOU WRITE

Would moving zero change the motion or only its coordinates? From ___ as zero, I predict ___ at time ___. I assumed ___.

# Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

## Reference frame

ES: sistema de referencia / PT: referencial

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain reference frame to a friend?<br>In my words, this means ___. | You report a cart at +5 m. What does a partner need to locate it?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| How could a seated passenger be at rest in one frame and moving in another?<br>My example shows ___. | Two people report different velocities. Does that alone prove one measured incorrectly?<br>I would change ___ because ___. |

## Components

ES: componentes / PT: componentes

| Meaning in my words | How can you tell? |
|---|---|
| What are the components of an arrow?<br>In my words, this means ___. | If east and north are positive, which component is negative for a southeast trip?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| What two arrows could you use to build a displacement of 3 m east and 4 m north?<br>My example shows ___. | A classmate counts the diagonal and both parts as three separate trips. What is counted twice?<br>I would change ___ because ___. |

# Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

## Units

ES: unidades / PT: unidades

| Meaning in my words | How can you tell? |
|---|---|
| What do units tell you about a number?<br>In my words, this means ___. | Before adding 2 m and 30 cm, what should you do?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| How would you add those two lengths in metres?<br>My example shows ___. | A partner adds the numbers and gets 32 m. What went wrong?<br>I would change ___ because ___. |

## Limit

ES: límite / PT: limite

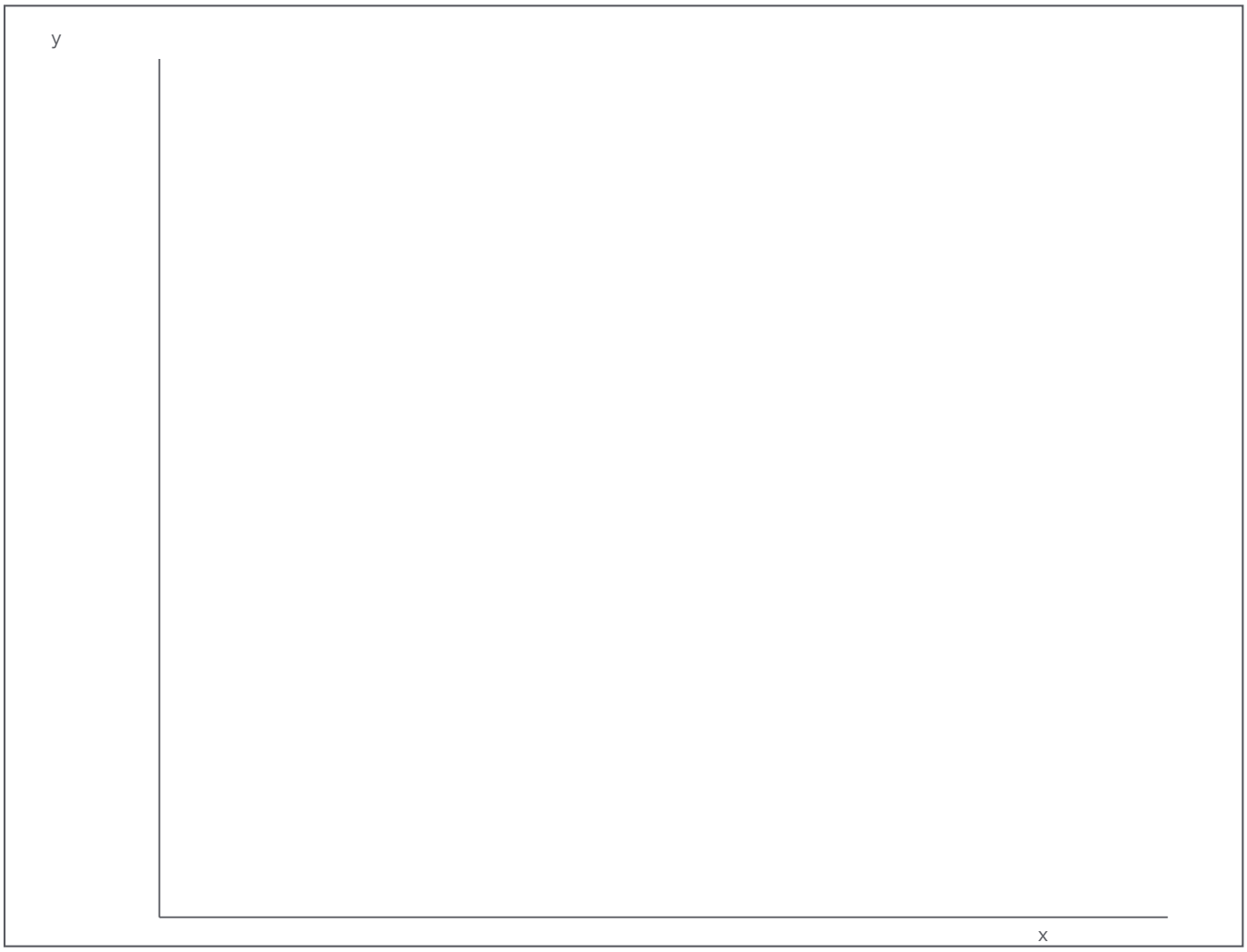
| Meaning in my words | How can you tell? |
|---|---|
| How would you explain limit to a friend?<br>In my words, this means ___. | Which change would make a steady-velocity forecast unreliable?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| What limit would you write next to a one-minute straight-line prediction?<br>My example shows ___. | A partner uses the one-minute model for a year. What evidence would they need first?<br>I would change ___ because ___. |

# Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

## Sketch the reference frame and one sideways component. Box your time unit and one model assumption.

Labels you can use: origin · direction · assumption.




**Tell a partner**

From ___ as zero, I predict ___ at time ___. I assumed ___.

# Two reports. One sighting.

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SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

**Our goal:** Compare position reports that use different zeros.

**CHOOSE ONE JOB TOGETHER**

**SURVEY ENGINEER** ☐

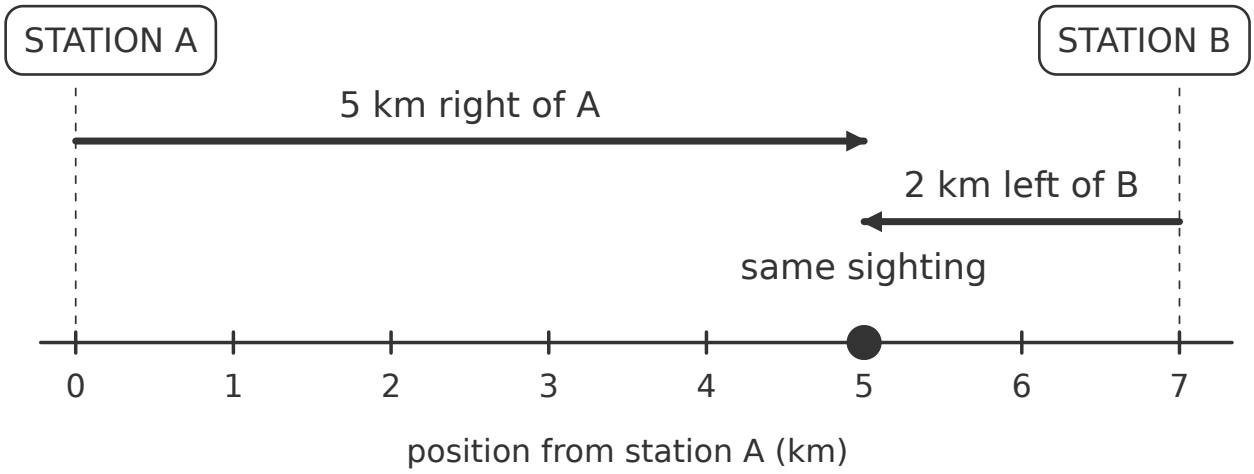
Use one axis for both stations. Give dispatch the position from A, then name what a future forecast still needs.

**MAP BROADCASTER** ☐

Show both stations and zeros on the locator map. Explain whether the two reports describe different locations.

**Word help:** zero / frame / position

Two observatories file the same sighting: "**5 km** right of station A" and "**2 km** left of station B." B is **7 km** right of A.




## 01 OUR FIRST CALL

**Do the reports agree? Give one location and explain how the map supports it.**

From station ___, the position is ___.

## 02 OUR REVISED CALL

**Wait for the reveal. Keep your first call.**

New clue: _____

What number changes in the new frame? What physical location stays the same?

Now we would say ___ because ___.

**Switch speakers.** What measurement uncertainty should accompany a future central position forecast? Give a concrete example.

| ☐ GREEN | ☐ RED | ☐ YELLOW | ☐ BLUE | ☐ PURPLE | ☐ ORANGE | ☐ PENCIL |
|---|---|---|---|---|---|---|
| speeding up | slowing down | steady / balanced | position / displacement | velocity | forces | other marks |